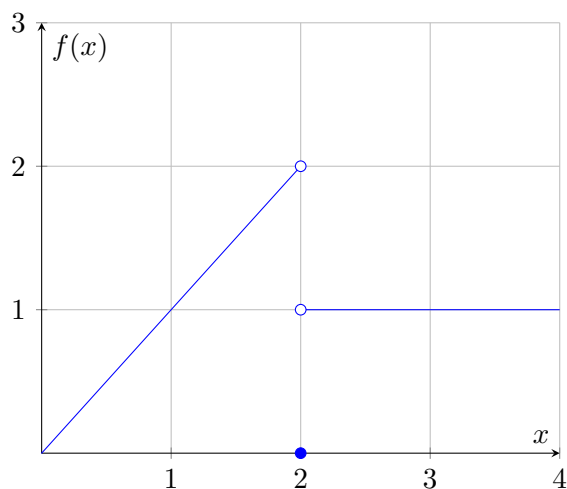


Section 10.1

1. Use the graph of the function f shown to estimate the indicated limits.



- (a) 5 points $\lim_{x \rightarrow 2^-} f(x)$
- (b) 5 points $\lim_{x \rightarrow 2} f(x)$
2. 10 points Find $\lim_{x \rightarrow -5} \frac{x+5}{x^2-25}$ or explain why the limit does not exist.

3. 10 points If $f(x) = \begin{cases} x + 5 & \text{if } x \leq 0 \\ 5 - x & \text{if } x > 0 \end{cases}$

Find $\lim_{x \rightarrow 0} f(x)$.

4. Given that $\lim_{x \rightarrow 2} f(x) = 9$ and $\lim_{x \rightarrow 2} g(x) = -11$, find the indicated limits.

(a) 5 points $\lim_{x \rightarrow 2} 2g(x) + 3f(x)$

(b) 5 points $\lim_{x \rightarrow 2} \frac{3x - f(x)}{x^2 + 1}$

5. A cell phone company charges \$60.00 for the first 1000 minutes of calls per month, and \$0.20 for each additional minute.

(a) 5 points (bonus) Write a piecewise definition of the charge $F(x)$ for x minutes of calls in one month.

(b) 5 points (bonus) Find $\lim_{x \rightarrow 1000}$ or explain why it does not exist. Show your work!

(c) 10 points (bonus) Graph $F(x)$. Label your graph, indicating the values of x and $F(x)$.

Section 10.3

1. 10 points Sketch a possible graph of a function f such that $\lim_{x \rightarrow 3^-} f(x) = 4$, $f(3) = 4$, and f is not continuous at $x = 4$.

2. Determine where each function is continuous. Express your answer in interval notation.

(a) 5 points $15x^2 + 2x$

(b) 5 points $\sqrt{15 - 3x}$ (Hint: think carefully about whether to include the point $x = 5$)

3. 10 points Let $f(x) = \begin{cases} \frac{x^2 + x - 6}{x - 2} & \text{if } x \neq 2 \\ 7 & \text{if } x = 2 \end{cases}$

Is $f(x)$ continuous at $x = 2$? Explain.

Section 10.4

1. Find the indicated quantities for $f(x) = 3x^2$.

(a) 5 points $\frac{f(2) - f(1)}{2 - 1}$

(b) 5 points $f'(1)$

2. The revenue (in dollars) from the sale of x tables is given by

$$R(x) = 150x - 0.15x^2, \quad 0 \leq x \leq 1000.$$

- (a) 5 points Find the average change in revenue per table if production is changed from 200 to 225 tables.

- (b) 10 points Find $R'(x)$ *by using the definition of the derivative* (i.e. by using limits, not the linearity of the derivative and the power rule).

- (c) 5 points Find the instantaneous rate of change of revenue at a production level of 200 tables.